

Blockchain

Blockchain tech is software

- ↳ a protocol for secure transfer of unique instance.
- ↳ No need of third party.
- ↳ Every node has equal control.

e.g : Money over IP

Application
Protocol

Email
SMTP

Phone calls
VoIP

Bitcoin
Blockchain

- Cryptocurrency is single application of Blockchain -

Ledger : Like record, i.e DB, spreadsheet etc.

- ↳ Blockchain is distributed network.
- ↳ No central node so no single point failure.

Blockchain is immutable ledger :

i.e. no delete, no revert, no update

once data is entered it will be permanent -

→ Cryptographic Identity :

Your own wallet address.

Web 1.0 = full form for identity

Web 2.0 = through email, github, etc.

Web 3.0 = through wallet address

Distributed Blockchain concept → **Satoshi Nakamoto**
Japanese
2008



First was Bitcoin blockchain
& coin used was Bitcoin.

دائری

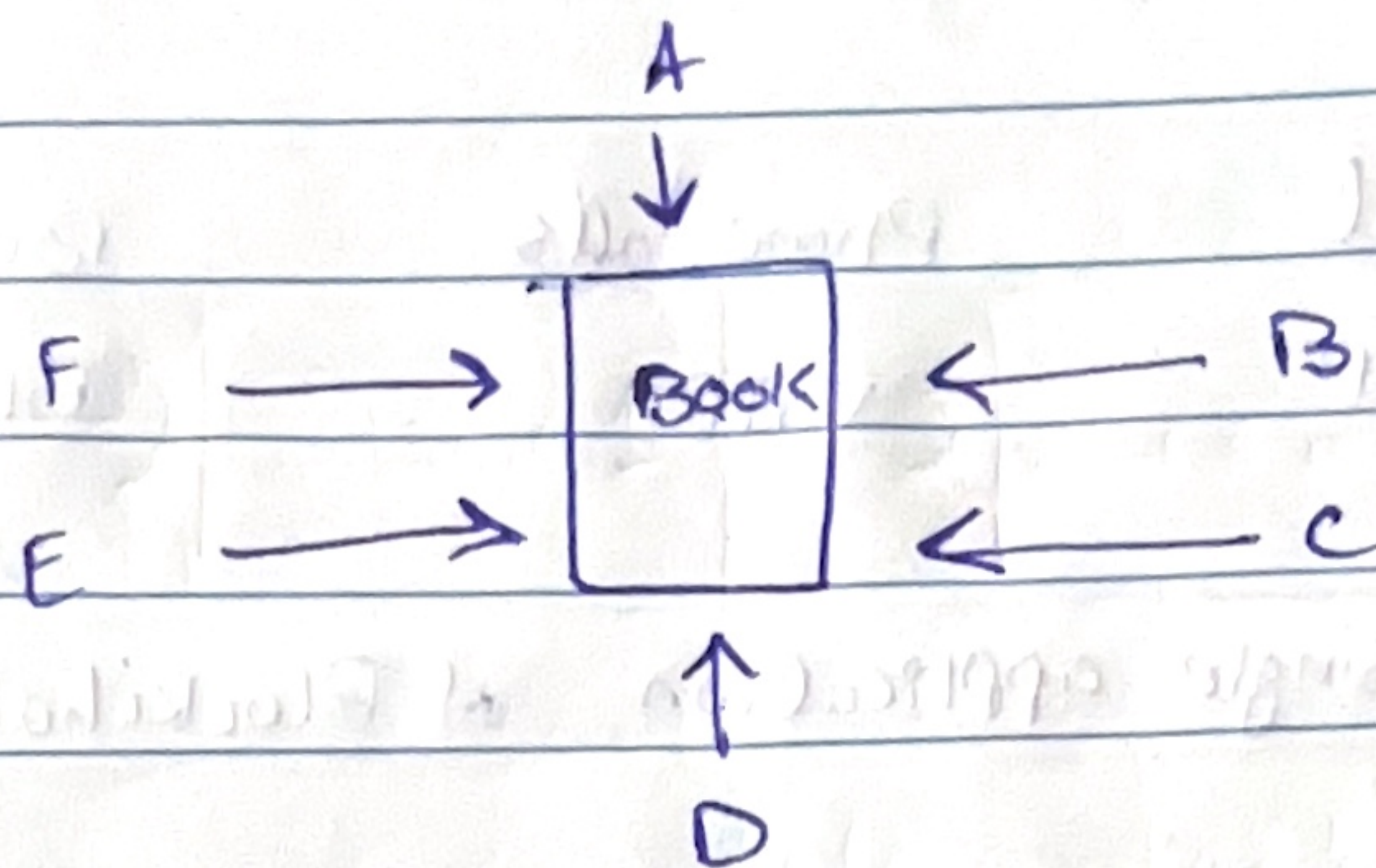
Consensus in distributed Networks:-

→ Yes / No

To win need at least ~~50%~~ 51% votes -

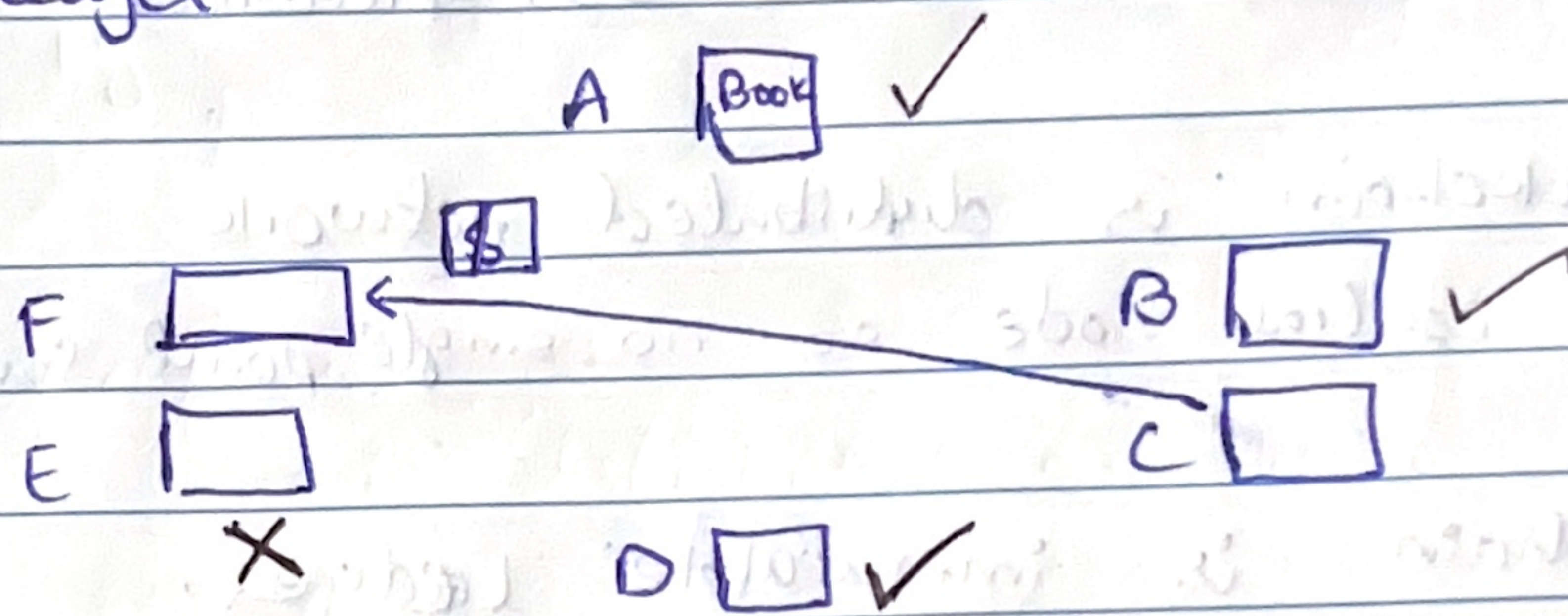
→ To execute transaction

Common ledger



A more Common ledger

Transaction C → F
executed as
majority validates.



★ → All transactions are public as we've to
validate through consensus.

AS Distributed ⇒ More secure.

Bitcoin Blockchain was
for financial transaction.

Evolution of Blockchain

2008 - 2013 : Bitcoin Blockchain 1.0 : Blockchain Emergence.

Bitcoin vs Blockchain

It's cryptocurrency Underlying tech
BTC is its coin. Cryptocurrency is its application.

2013 - 2015 : Blockchain 2.0 : Ethereum Development.

Added smart contracts with financial transaction

↳ Digital contracts.

2016 DAO attack (Decentralised Autonomous Organisation)

Ethereum 1.0

Ethereum 2.0

2020 → stable coins.

Definition:

↳ Decentralized

↳ Can be Public or Private

↳ Stored data is immutable

↳ Highly secured

↳ Data gets recorded via consensus based algo.

↳ Use Cryptography

↳ Generally exists over Peer to Peer networks.

Each computer participates in P2P is called Node-

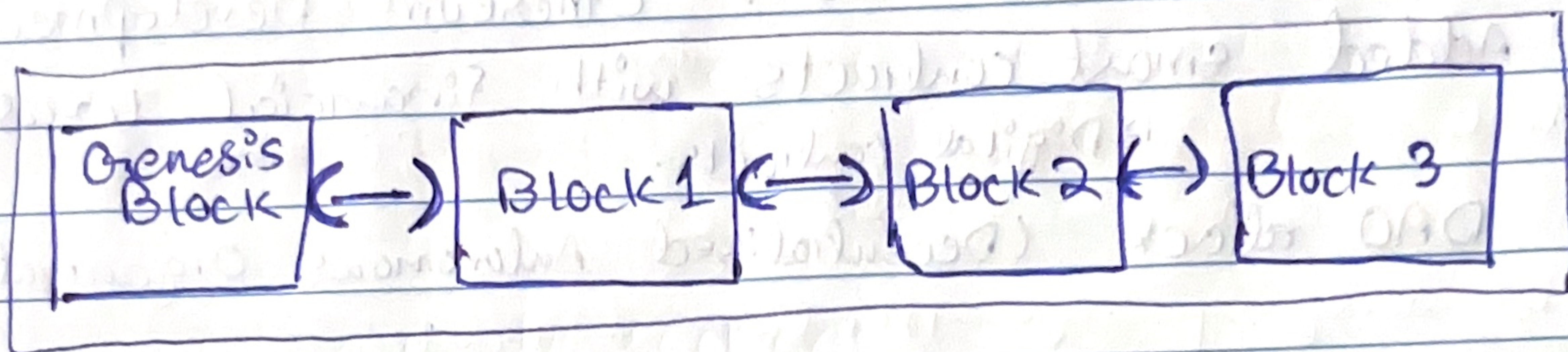
Each node maintains the records of transactions in

multiple consecutive blocks

Blockchain tech is called Decentralized Ledger Technology (DLT)

Ledger Technology (DLT)

as each node keeps same copy of ledgers.



- user is anonymous while blockchain is public.
 - ↳ my identity
 - ↳ our wallet address.

Properties of Blockchain:

↳ ★ Both Decentralized & distributed networks

- * Distributed ledger.

- ★ Fault Tolerance (no failure are not centralized)

- ★ Attack Resistance (51% favorable need)

- ★ Remove Intermediaries (No 3rd Party between)

• $\rightarrow$ for the people by the people -

- * Consensus Protocol (a specific algo.)

e.g. PoW → Proof of work (Bitcoin uses this)

e.g. PoS → Proof of Stake (also mining)

(etherium now uses this)

first on POW now
on PAS

In 2022 Pow to Pos (Ethereum)

- ★ Faster Settlement (Faster trans. b/w 2 parties as no 3rd party)
- ★ Low Transaction Fee (0.02% to 0.04%)
- ★ Transparency (all trans. are available publically)
- ★ Immutability (Settlement transaction in case of wrong)
- ★ Irreversible transaction
- ★ Trustless System (trust is on Blockchain not on user)
- ★ Availability (As no single point of failure so others are available)
- ★ Empowers Individual (each control its own wallet)
- ★ Chronological order of transaction (sequence is maintained)
- ★ Timestamped (Time is marked)
- ★ Sealed with cryptography (use hashing algos, Asymm etc.)

Byzantine fault tolerance

Bit coin's Monetary Policy:

Reward will be halved after 4 years

The Halving
(Reward Halve Every 4 years)

Block Frequency
(on Avg each block is mine in 10 min on Avg)

In Bitcoin Block chain all trans. are in MemPool

~~After 2020~~ Reward was 6.25 btc coin
2020-2024 Reward was 12.5 btc coin
2016-2020 Reward was 25 btc coin
2012-2016

All miners competes in mining the block -
Reward is given - 3.125 bitcoin in
minning one block -

Each block is minned in 10 mins on avg - $\Rightarrow$ Block frequency.
All machines try to mine single block & they
complete -

ASIC machines $\rightarrow$ Specific for Mining -
 $\rightarrow$ Application Specific Integrated Circuit -

In block we've transactions

Every data is stated as transaction

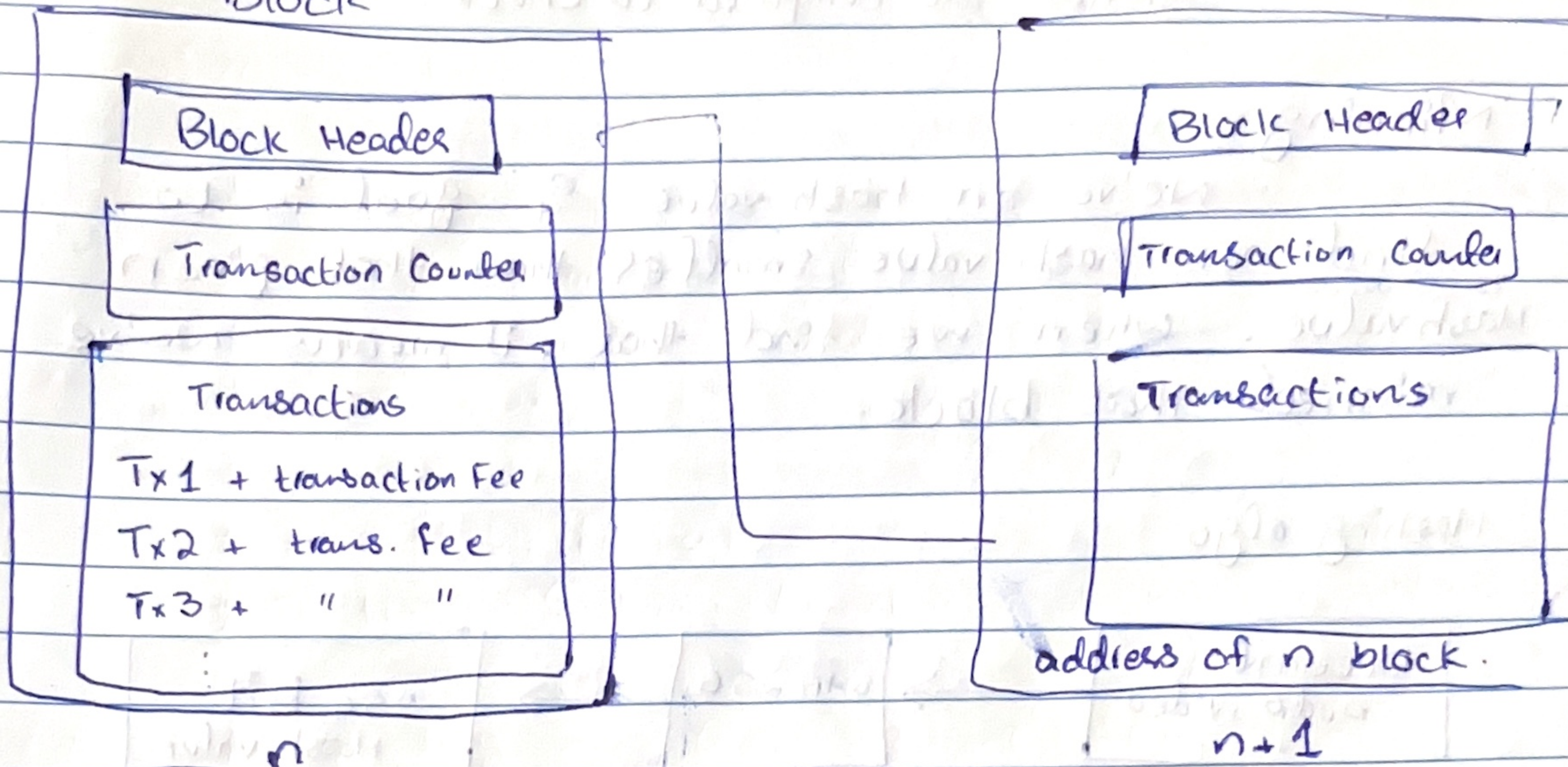
Block's Properties

- Header
- Previous Block Address / Hash
- Timestamp
- Nonce : Backbone of POW algo. i.e Nonce value.
- Merkle Root : Data structure childs of Markel Tree are the hash of transactions used for verification purposes.

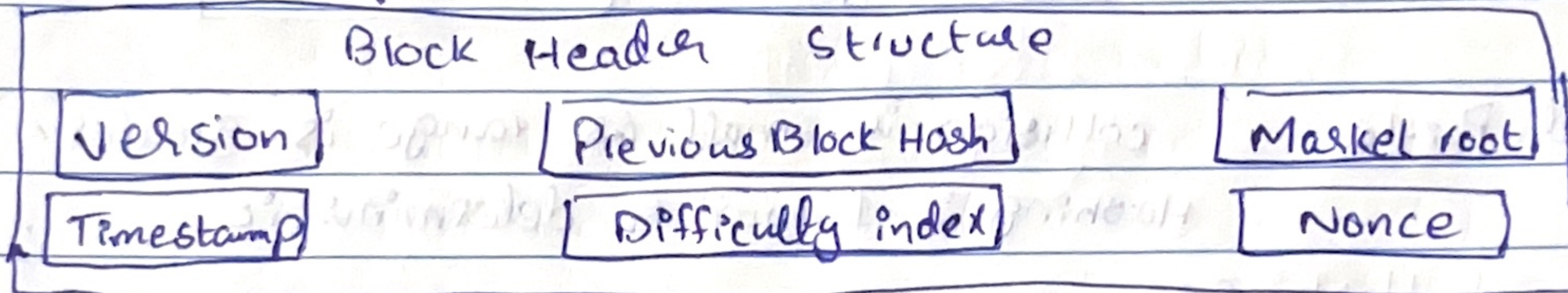
Types of Blocks

- Genesis Block : 1st Block of blockchain
- Valid Block : valid which can be added to blockchain.
- Orphan Block : Not valid, i.e which was minned (Uncle Block) but not be part of blockchain as only one block at that time -

Block



Block Header Structure



Nonce is 32 bit number

$\Rightarrow 2^{32}$ numbers possible.

The Nonce

Hashing is used for data integrity

Minning is Mathematical puzzle
to solve.

Blockchain gives value
of Hash & ^{goal} result is to
calculate ^{Hash value} small value than that
Hash value.

Hash is
calculated from
these values

Block No: 6
Nonce
Data:
KSEJ → Rakesh 500 coins
Raj → Bella 200 coins
Prev Hash: 000AB23
Hash:

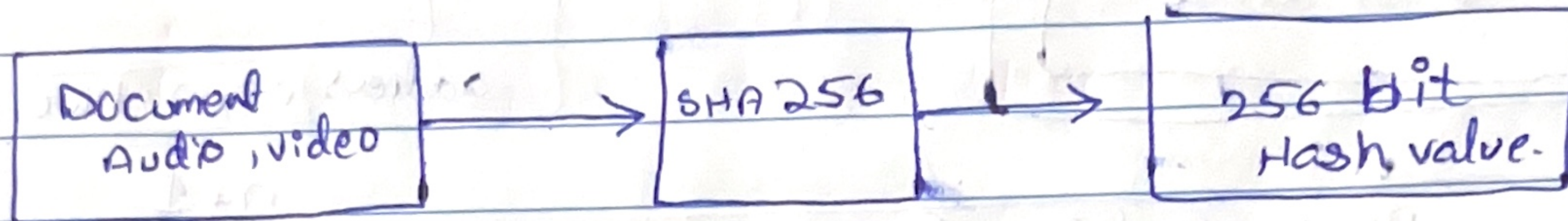
Fixed sized
Hash i.e 256 bits

Hashes are compared to check.

Minning:

We've an hash value & goal is to find new hash value smaller than that given ^{target value} Hash value. When we find that, it means we've minned that block.

Hashing algo



Crossword Puzzle

collusion is small as range is greater.
Hashing is 1 way & deterministic

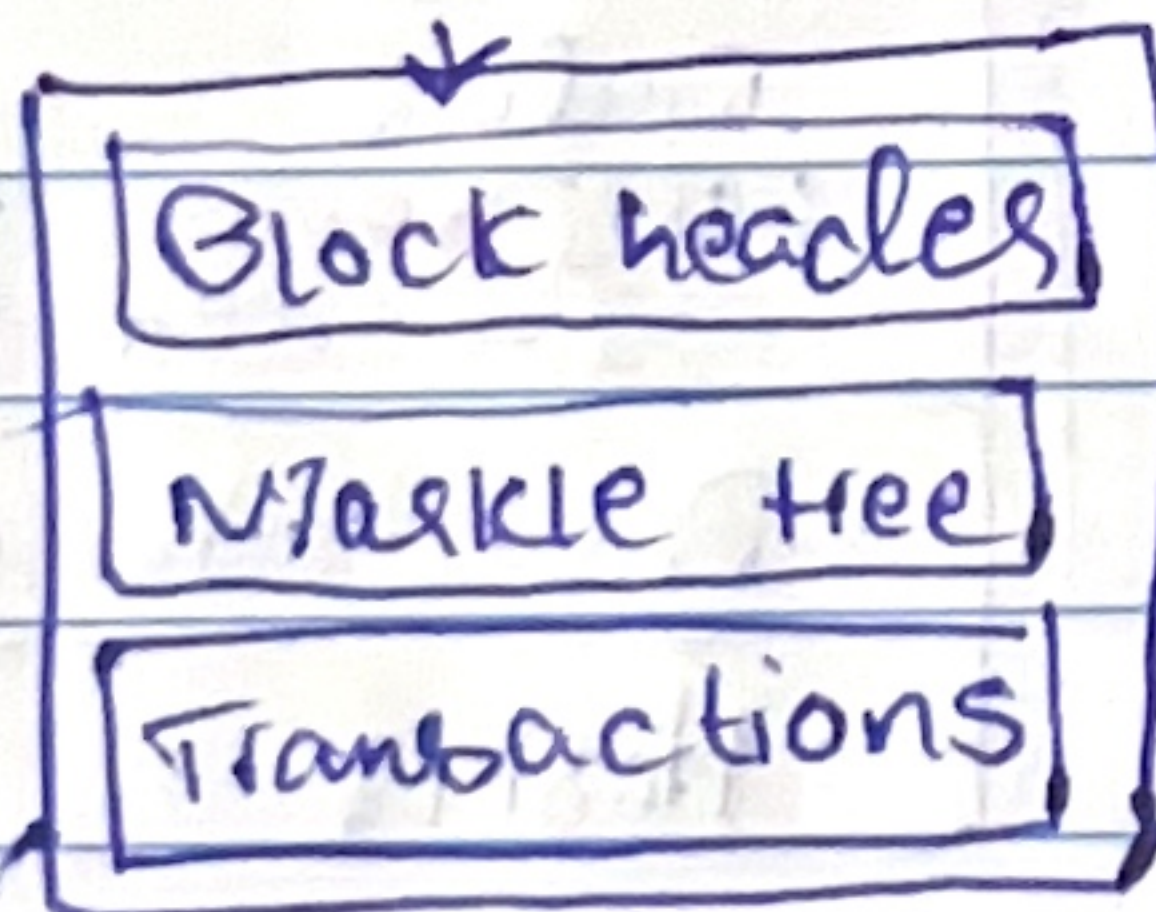
Benefits of Hashing:

- Reduce Size
- Time Saving

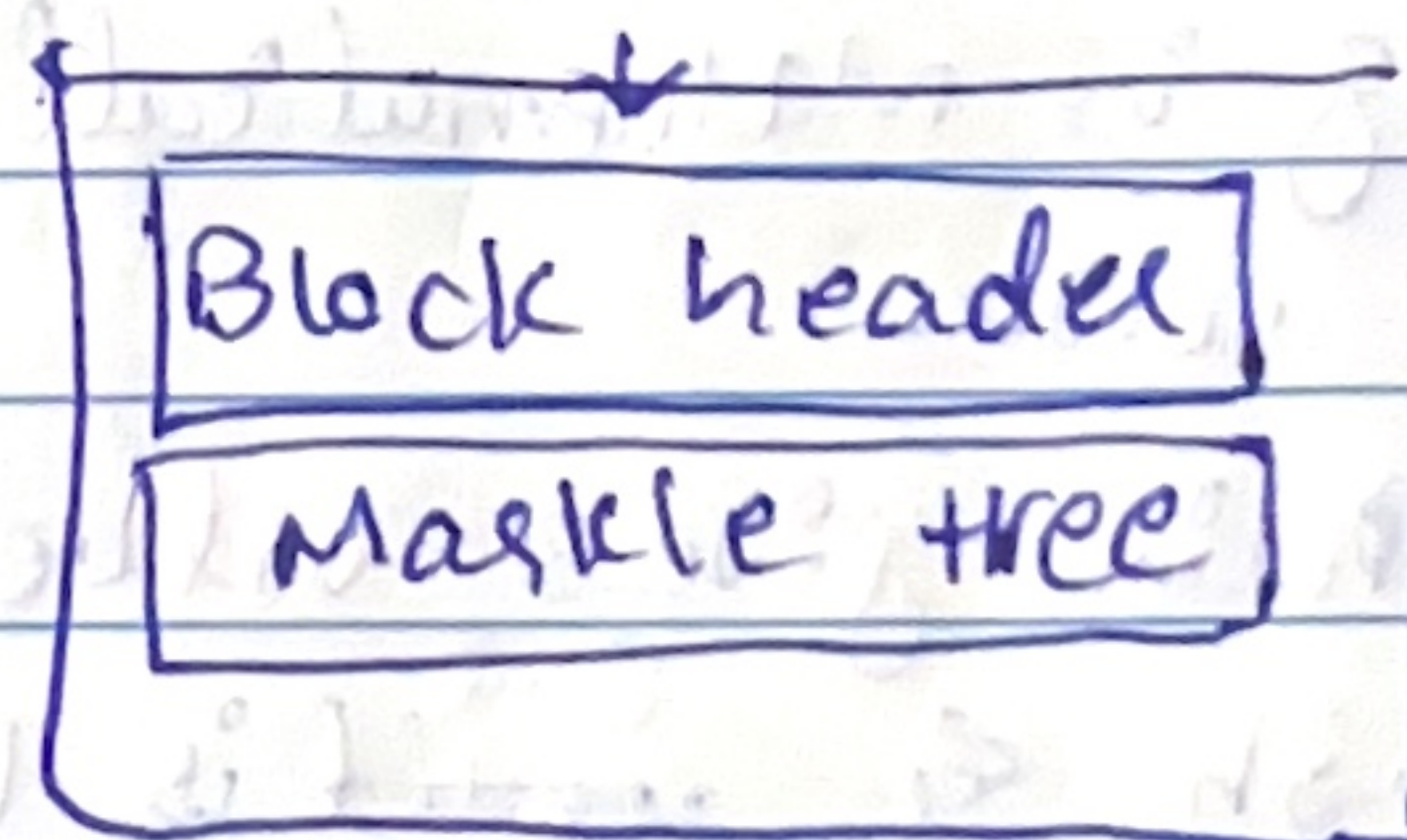
Evalaunch effect: small change has very large effect.

Types of Nodes:

• Full Node



• SPV Node



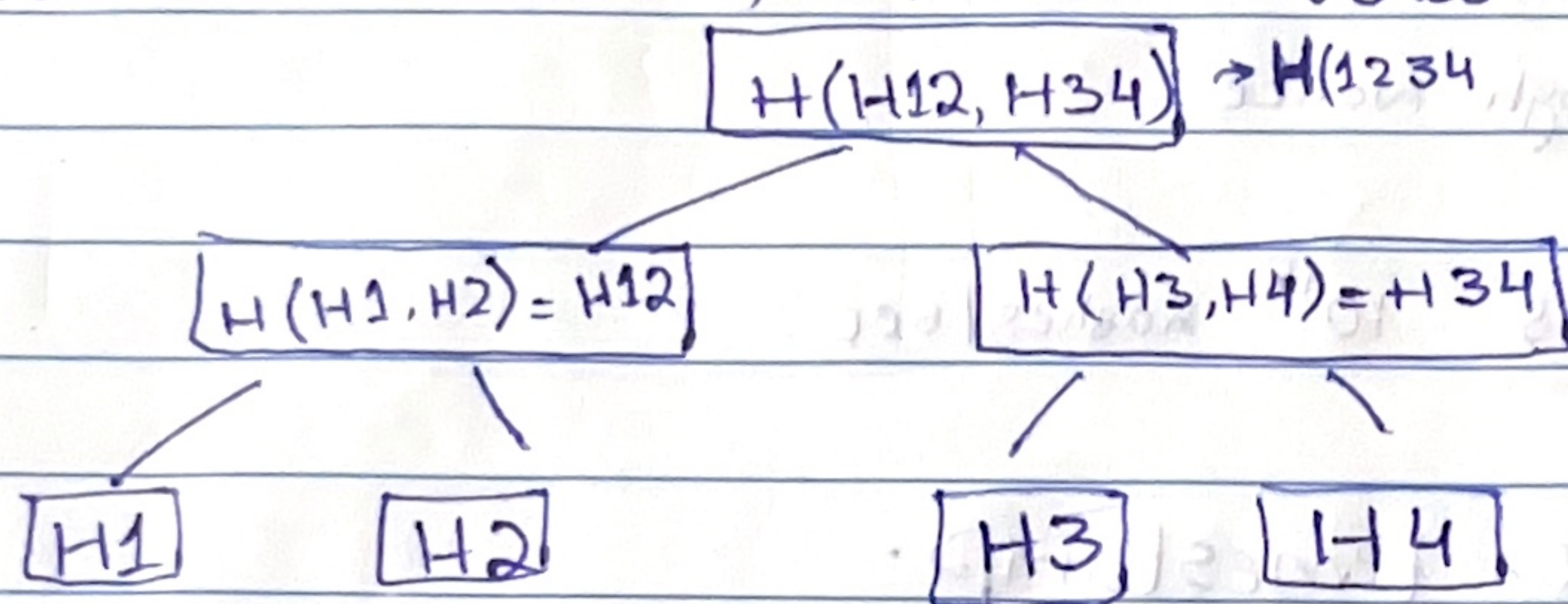
for verification Purposes.

Merkle Tree is also known as Hash tree

↳ Used for Data Verification Not for Data Searching

↳ Every block has its merkle tree as per their own transactions -

Let $H1, H2, H3, H4$ are Hashes of $T1, T2, T3, T4$



Target: Less Hash value ~~sho~~ than target, should be found -

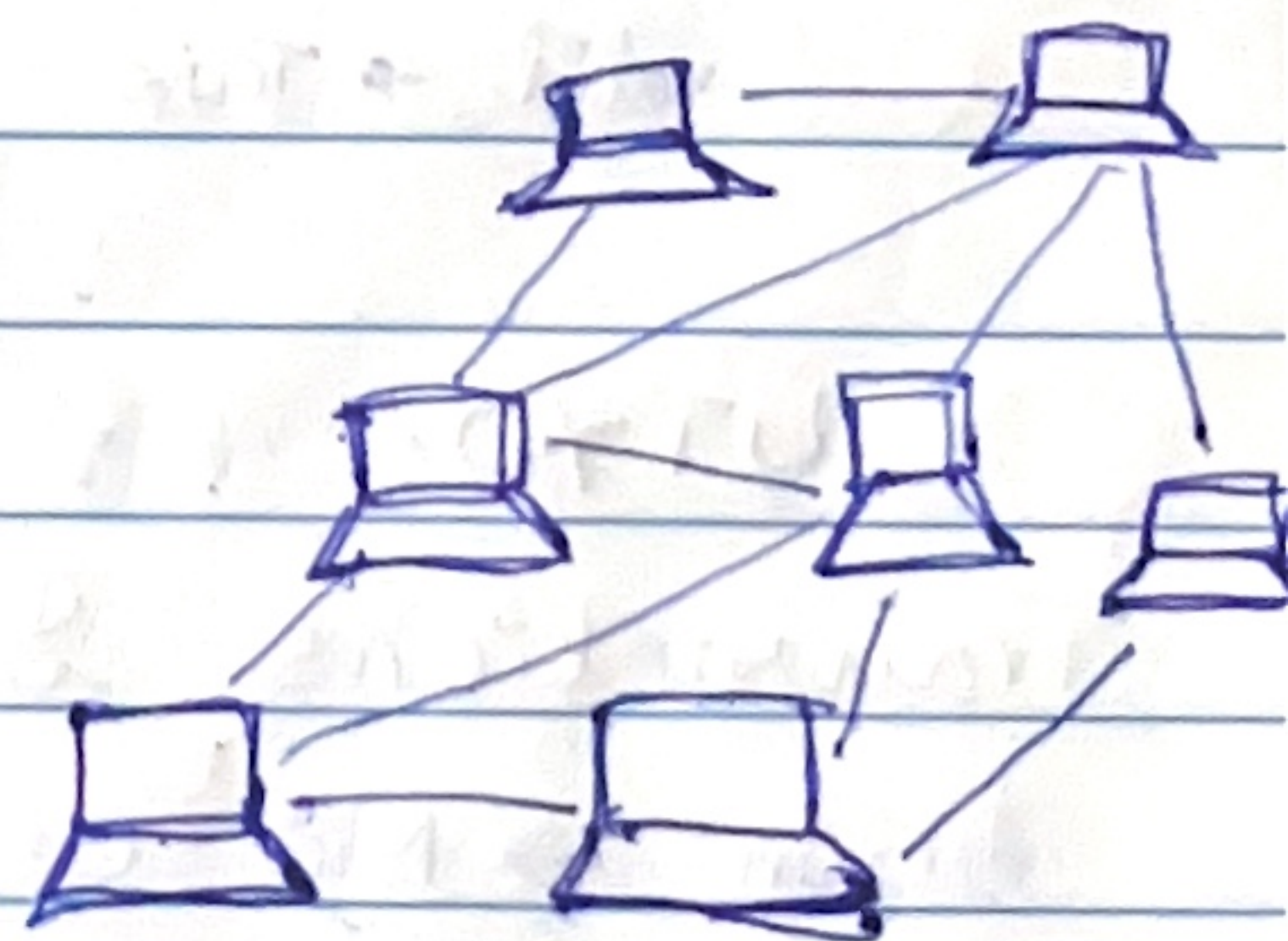
Target adjusts every 2016 blocks
roughly 2 weeks.

Ensure block is mined in avg 10 mins

Mining Pools:-

ASIC Machines Grouping -

Rewards is also shared - Depending on Hash power



Bit Coin White Paper.

Nonce Range:

32 Bit Number

Range 0 to $2^{32} - 1$

Hash size is 256 bits

$\Rightarrow 0$ to $2^{256} - 1$

total valid Hashes $\approx 10^{77}$

total Nonce $\approx 4 \times 10^9$

These are not enough Nonce

A modest mines does 10^{18} hashes/sec.

Time Stamp:-

UNIX timestamp.

Any number at any second.

- Data Change
- Timestamp change.

6 Confirmations:- 6 blocks added after a block

After 6 confirmation miner gets reward - after this

Valid $\rightarrow$ True

OPPhan $\rightarrow$ False

transaction is irreversible

UTXOs : unspent Transactions Output.

Transactions & UTXOs $\rightarrow$ unused transactions.

Arjun $\rightarrow$ Me 0.4

Raj $\rightarrow$ Me 0.3

Alice $\rightarrow$ Me 0.7

Bob $\rightarrow$ Me 0.1

0.15

UTXOs

Transaction:

input

0.7 BTC from Alice

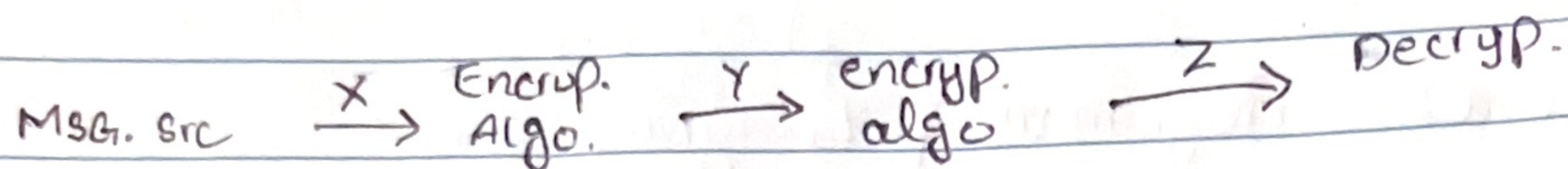
output

0.5 BTC to coffee shop

0.2 BTC back to me

I'll say I buy coffee for 0.5 BTC

Coin $\rightarrow$ Native currency
Blockchain's own currency
Token $\rightarrow$ Launch someone's blockchain
currency -



Wallet:- $\rightarrow$ is basically a software who manage your account.

Piece of code

Secure cryptography wallet.

store coins & tokens

Holds keys -

Can change wallet easily -

Types :

Cold wallets

For storing cryptocurrency for a longer term.

Offline wallets

Hot wallets.

Mobile wallet

Desktop wallet

web Based wallet

MetaMask is software.
You can make Multiple
wallets in it -



Networks

Main net:
Testnet: During Development
↳ Free.

It's a good practice not to store complete
currency in single wallet.

Control & Responsibility

Smart Contract → Digital Code

Bitcoin Script

↓
Not Turing complete
(not General Purpose Lang.)

SOL

(specific Purpose)

→ Lang. for Ether.
smart code

Solidity

↓
Turing Complete
(General Purpose Lang.)

C, C++, ...

Solidity is Lang for Smart Contract
of Ethereum
Extension is .sol.

→ (updating smart contract)

Ques: How bug can be fixed in smart contract? or
update

Ans: We've to deploy updated contract -

EVM: Ethereum Virtual Machine

↳ sandbox - ↳ smart ~~code~~ Contract are kept there
for security purposes -

Remix IDE for testing purpose of deployment of Ethereum
smart contract.

→ Ethereum is Open Source -